NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

FSCS – Midterm Examinations Spring 2025

Time: 90 minutes

CT162 – Discrete Structures (Paper B)

Max Marks: 20

Note: Attempt all questions.

Q1: Determine whether the following function is a bijection from R to R.

$$f(x) = (x^2 + 1)/(x^2 + 2)$$

[CLO-1 3 marks]

Q2: Find the value of $\sum_{j=0}^{8}(3 - (-1)^j)$

[CLO-1 3 marks]

Q3 Let $U = \{x \mid x$ is a positive integer less than 9$\}$, $A = \{x \mid x$ is an even positive integer less than 7$\}$ & $B = \{x \mid x$ is an odd positive integer less than 5$\}$, Find (i) $A \cap \overline{B}$, (ii) Power Set of B

A = {2, 4, 6}

AU = {1, 2, 3, 4, 5, 6, 7, 8}

[CLO-1 2 marks]

Q4: For the given argument, give its argument form and state which rule of inference is used.

"We will go swimming when it gets very hot. A sufficient condition for going out for ice cream is we go swimming. Therefore, if it gets very hot, then we will go out for ice cream."

[CLO-2 3 marks]

Q5: Are these system specifications consistent?

"The system is in multiuser state if and only if it is operating normally. If the system is operating normally, the kernel is functioning. The kernel is not functioning or the system is in interrupt mode. If the system is not in multiuser state, then it is in interrupt mode. The system is not in interrupt mode."

[CLO-2 3 marks]

Q6: Use the bubble sort to sort [d, f, k, m, a, b] showing the lists obtained at each step. Also, mention what is the complexity of this algorithm.

[CLO-3 3 marks]

Q7: Explain the working of insertion sort algorithm and also mention its worst time complexity.

[CLO-3 3 marks]